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Project P8

Enhancing Web Accessibility Standards

*Implementation of User-Centric Customization Interfaces and
CSS-based Contrast Optimization*

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1. Abstract

[TBD]

2. Acknowledgement

[TBD]

3. Introduction

The previous project, *Accessibility on the Web – Where we Stand* [1], examined the fundamental principles of digital inclusion in modern web applications. It explored how semantic HTML forms the foundation of accessibility, provided an overview of both common and lesser-known native elements, and clarified the roles of Accessible Rich Internet Applications (ARIA) and Web Content Accessibility Guidelines (WCAG). In addition, it introduced the basics of accessibility debugging, testing, and the practical use of screen readers.

Building on that foundation, this project focuses on essential aspects of accessibility that may appear less technical in the traditional sense, but are equally important for creating an inclusive digital world. While the previous project concentrated primarily on the structural and technical mechanisms behind the scenes, it did not examine the visual design and the resulting contrasts of web applications. These design decisions are far more than aesthetic choices. They play a decisive role in determining whether a web application becomes a welcoming space or a digital dead end for many users.

3.1. Objective of this Project

This project aims to explore accessibility in greater depth by examining ways to build on and extend existing browser standards, improving readability by optimizing contrast. It also investigates additional techniques and approaches that can simplify the user journeys of people with impairments, making web applications more user-friendly and adaptable to individual needs.

3.2. Research and Ideation

The initial direction of this project explored how accessibility can be tested in an automated manner. During this research phase, a wide range of accessibility testing tools was examined, including both free and commercial solutions. One particularly influential source was the blog of Karl Groves, which states that only around 30–40% of accessibility issues can be reliably detected through automated testing [2]. While automated accessibility testing has improved since the article was first written, the research process gradually shifted the focus of this project into another direction.

Additional ideas emerged regarding accessibility support beyond testing itself. Instead of focusing exclusively on automated evaluation, the project evolved toward exploring ways to enhance browser standards and provide additional accessibility aids considering contrast optimization and application customization directly within web applications. Since accessibility testing is already heavily researched and commercially established, this alternative direction appeared more valuable and interesting for further exploration.

An exploratory prototyping approach was chosen to rapidly test and refine ideas over time. Existing standards, emerging browser capabilities, and newly introduced CSS

and JavaScript features were researched to build a broad overview of current possibilities.

Browser support and compatibility were considered throughout the process to better understand both practical limitations and future opportunities. Experimental features and unconventional approaches were intentionally included where appropriate to avoid building solutions solely around already outdated technologies and standards.

4. Auditing Existing Applications

The first part sets its focus on existing web applications. It introduces key concepts of contrast perception, how current standards are defined, and in what direction they are evolving.

Based on this, tools are introduced which were created during this project. A color contrast checker is provided, enabling the calculation of WCAG 2.x-based contrast ratios between two colors using relative luminance. Additionally, a Developer Widget is provided which simulates different vision impairments. This allows inspection of websites under simulated conditions to evaluate their current accessibility state with respect to contrast and to support the identification and improvement of potential weaknesses.

4.1. Setup and Exploratory Prototyping

The starting point of this project included the creation of an isolated environment, referred to as the Sandbox project, for rapid prototyping and experimentation. This environment allowed different implementation approaches to be tested incrementally while individual project parts evolved over time.

Testing CSS and JavaScript compatibility directly in practice proved especially important, as documentation alone often did not fully reflect actual browser behavior. Experimentation and exploratory iteration became a central part of the ideation process before more concrete concepts gradually emerged and were refined into the implementations presented throughout this documentation.

Many examples initially started as simple or partially hacky drafts before being iteratively improved into more stable and reusable solutions.

4.2. Mathematical Foundation

To understand the current WCAG 2.x contrast standard, this work analyzes the W3C documentation regarding relative luminance and RGB color calculations. This defines the mathematical relationships between RGB values, luminance calculation, and contrast ratios.

This understanding allows the algorithmic approach from the specification to be formalized into a clear mathematical representation. This formulation enables the implementation of the contrast ratio indicator used within the custom contrast color function application.

This analysis also motivated research into the upcoming WCAG 3.0 contrast model. The WCAG 2.x documentation itself acknowledges limitations in its current formula, including simplifications, rounding inaccuracies, and perceptual shortcomings. While the APCA model was identified as a promising future replacement, it was not integrated into the practical implementation of this project because it remains under active development in draft status.

4.3. Contrast Perception

Contrast is essential for distinguishing elements and understanding their relationships. Differences in brightness, size, sharpness, and shape all contribute to visual contrast. On the web, variations in lightness are the primary means of ensuring text readability and clearly separating interface elements.

The following section introduces the contrast requirements defined in WCAG 2.x and provides an overview of the emerging approach proposed for WCAG 3.0.

4.3.1. Relative Luminance

The contrast requirements in WCAG 2.x are based on the relative luminance of text and its background, independent of hue. This approach assumes that hue and saturation have little influence on reading performance, even for users with color vision deficiencies. Since the inability to distinguish certain colors does not usually affect the perception of light and dark, color itself is not treated as a primary factor in the contrast calculation.

In practice, the perceived contrast may differ from the theoretical value. Smaller or thinner fonts can appear noticeably lighter than the color specified in CSS due to anti-aliasing and font rendering. As a result, text may appear to have lower contrast than the calculated ratio suggests.

WCAG 2.x requires a minimum contrast ratio of 4.5:1 for normal text to meet Level AA. Large text must reach at least 3:1. For Level AAA, the minimum contrast ratio increases to 7:1. These thresholds are based in part on recommendations from ISO 9241-3 and are intended to compensate for reduced contrast sensitivity in users with moderate visual impairments. Users with more severe vision loss typically rely on assistive technologies such as screen magnifiers or screen readers.

Color vision deficiencies vary significantly, making it difficult to define universally effective color combinations based solely on hue. This is one of the main reasons why WCAG evaluates contrast primarily through relative luminance.

One notable exception is **protanopia**, in which red tones may appear significantly darker than expected. As a result, red text on dark backgrounds can provide much less contrast than the calculated ratio suggests and should generally be avoided. [3]

The formula shown in Listing 1 is used by WCAG 2.x to calculate relative luminance, which serves as the basis for determining contrast ratios. [4]

$$L = 0.2126 \cdot R + 0.7152 \cdot G + 0.0722 \cdot B$$

$$\text{Where } C = \begin{cases} \frac{C_{\text{sRGB}}}{12.92} & \text{if } C_{\text{sRGB}} \leq 0.03928 \\ \left(\frac{C_{\text{sRGB}} + 0.055}{1.055} \right)^{2.4} & \text{otherwise} \end{cases}$$

Example: Dodger Blue `rgb(30, 144, 255)`

1. Normalize (/255)

$$R_{\text{sRGB}} = \frac{30}{255} = 0.1176$$

$$G_{\text{sRGB}} = \frac{144}{255} = 0.5647$$

$$B_{\text{sRGB}} = \frac{255}{255} = 1.0000$$

2. Linearize

$$R = 0.0123$$

$$G = 0.2747$$

$$B = 1.0000$$

3. Calculate Luminance (L)

$$L = (0.2126 \cdot 0.0123) + (0.7152 \cdot 0.2747) + (0.0722 \cdot 1.0000) = \mathbf{0.2712}$$

Calculate Contrast Ratio

$$\text{Ratio} = \frac{L1 + 0.05}{L2 + 0.05}$$

Listing 1: Step-by-step calculation of relative luminance based on the W3C sRGB formula. [4]

4.3.2. Perceptual Color Models

The contrast model used in WCAG 2.x is based on relative luminance. In the future, it is expected to be replaced by the Accessible Perceptual Contrast Algorithm (APCA), which forms part of the ongoing WCAG 3.0 work. Advances in display technology, modern web design, and vision science have shown that the approach introduced nearly two decades ago in WCAG 2.x no longer reflects how contrast is perceived in practice.

WCAG 2.x follows a binary model in which a contrast ratio either passes or fails. APCA takes a different approach by accounting for the non-linear nature of human perception and by evaluating contrast in the context in which the colors are used.

Readability depends on more than the luminance difference between two colors. Font size, font weight, stroke thickness, and surrounding context all influence how contrast is perceived. High light-dark contrast generally improves readability, but smaller and thinner text requires a greater difference in lightness to remain legible, as illustrated in Figure 1.

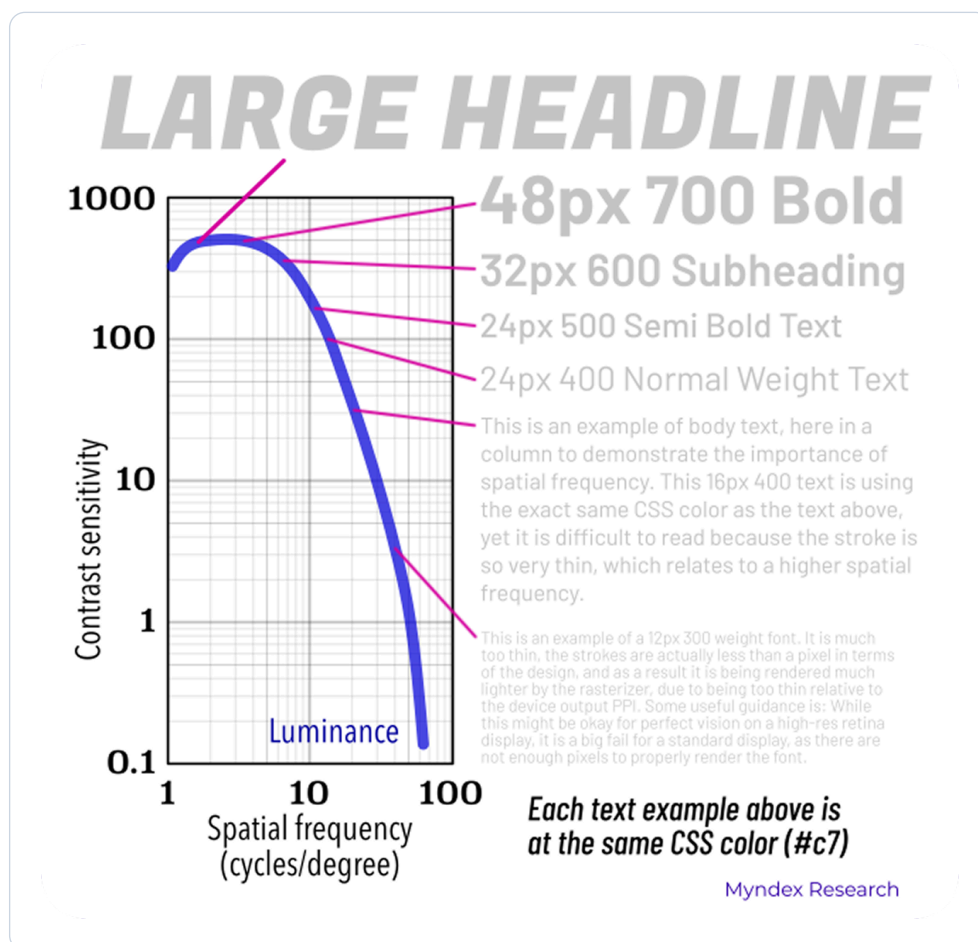


Figure 1: This chart illustrates the spatial dependence of human contrast sensitivity using text samples. [5]

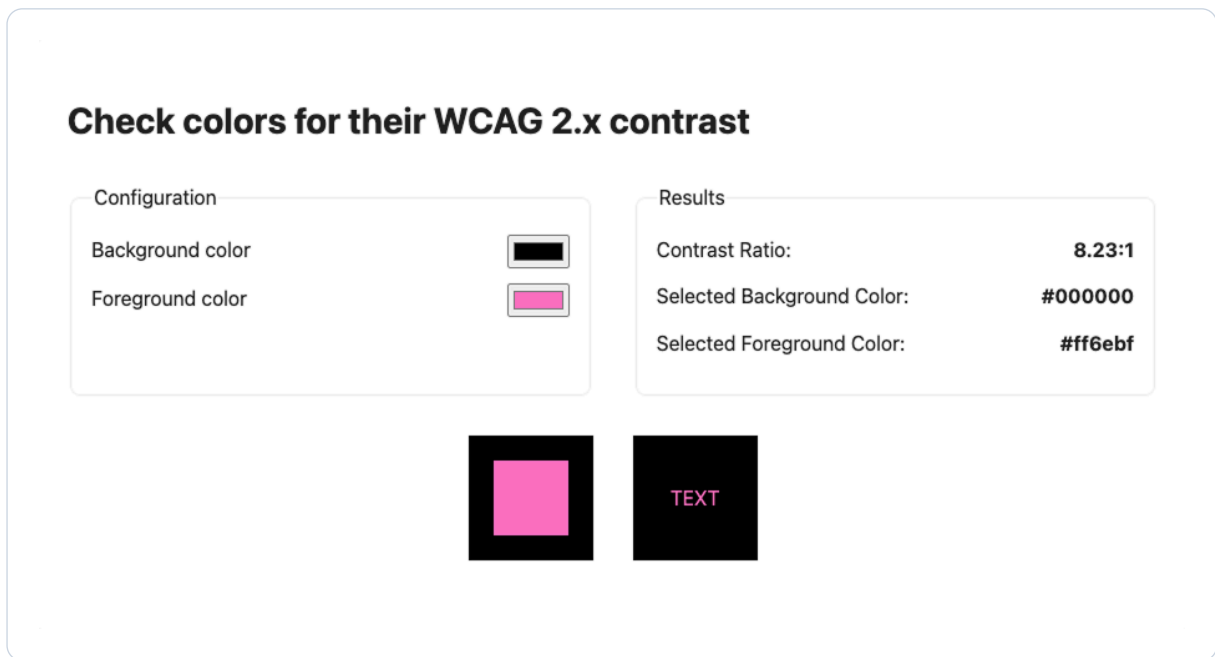
Non-text elements such as icons typically require less contrast than body text. More generally, the appropriate contrast between two colors depends on the specific use case, including the size, thickness, and purpose of the element.

The contrast formula used in WCAG 2.x has been criticized for many years. Case studies comparing white and black text on colored backgrounds have shown that combinations classified as inaccessible under WCAG 2.x were sometimes perceived as more readable by participants with color vision deficiencies. [6]

APCA introduces a new metric called lightness contrast, expressed as L^c . Instead of assigning a universal pass-or-fail threshold, it estimates readability based on the perceptual relationship between foreground and background colors within their actual visual context. [5]

4.4. Color Contrast Checker

The WCAG contrast ratio requirements are precisely defined but cannot be reliably estimated by visually comparing two colors. Therefore, additional tools are required to verify whether these requirements are fulfilled for a given color pair. This is where the Color Contrast Checker, shown in Figure 2, comes into play. This tool allows the selection of two colors and calculates the contrast ratio to verify whether a color combination is suitable for use in a web application or whether alternative colors are required to ensure sufficient contrast.



The screenshot shows a web application titled "Check colors for their WCAG 2.x contrast". It is divided into two main sections: "Configuration" and "Results".

Configuration:

- Background color:** A color picker showing a black square.
- Foreground color:** A color picker showing a pink square.

Results:

- Contrast Ratio:** 8.23:1
- Selected Background Color:** #000000
- Selected Foreground Color:** #ff6ebf

Below the configuration and results sections, there are two visual representations of the color pair:

- A black square with a pink square inside it.
- A black square with the word "TEXT" in pink.

Figure 2: A screenshot of the color contrast checker that calculates the ratio of two selected colors.

The implementation of the color contrast checker is available in the project's repository. [7]

4.5. Developer Widget

While having a design with colors coordinated with each other is important, it is also essential to consider the system as a whole. All individual elements incorporated into a webpage must harmonize with each other in a consistent visual structure. The Developer Widget shown in Figure 3 is intended to visually evaluate how a website behaves under different vision impairments. In its current state, this widget allows simulation of common color blindness types in combination with blurred vision, and can be extended to a wider range of accessibility conditions if required.



Figure 3: A screenshot of the developer widget where a website is inspected simulating Protanopia combined with intense blurry vision.

The widget itself is a simplified version of the Preferences Widget introduced in a later chapter. Implementation details are discussed in more depth later, as this widget shares the same underlying mechanism but operates with a reduced feature set. It applies filter configurations by adding classes to the document body, based on bookmarklets developed in the previous P7 iteration. [8]

While the simplified Developer Widget modifies the document state through direct class mapping, the more advanced architecture of the full Preferences Widget, detailed in Section 6.3, manages these settings globally by dynamically binding values to CSS custom properties on the root element.

The implementation of the Developer Widget, alongside an example page containing various content types, is available in the project’s repository. [9]

5. Guiding Developers During Construction

The following section discusses aspects to consider while building a web application and introduces tools that support developers throughout the development process. These tools make it easier to address contrast concerns early, resulting in more accessible websites.

A custom CSS contrast color function is presented that extends the functionality currently available in CSS. Additionally, an interactive Ishihara plate is introduced to evaluate the color compatibility of a design while also allowing its colors to be inspected under simulated color blindness conditions. The section further covers practical CSS and JavaScript concepts, demonstrating how developers can respond to user preferences and implement adaptive behavior using CSS custom properties.

5.1. Color Models

There are many different models for describing the colors perceptible to the human eye. Each model has its own strengths and weaknesses and is used in different contexts, ranging from print and photography to digital displays.

For many years, CSS relied primarily on the RGB color model, which limited how colors could be described and manipulated. To update features faster, the W3C now evolves CSS through independent modules rather than giant versions. Under this system, the CSS Color Module Level 4 specification acts as a direct software update to the three previously released levels, introducing support for additional color models and more advanced color functions to significantly expand the possibilities for working with color in CSS. Development in this area continues, and the W3C is currently working on the CSS Color Module Level 5 specification.

5.1.1. RGB

The RGB color model represents colors by combining different intensities of the primary colors `red`, `green`, and `blue`. An optional alpha channel can be added to define the opacity of the resulting color. The geometric relationship of these three primary channels is mapped as a three-dimensional cube, as shown in Figure 4.

RGB has been supported in CSS since its early days and can be expressed using the `rgb()` function or hexadecimal notation. For example, the color shown in Listing 2 can also be written as `#2d1fb180`. [10]

This color model is widely used because it has been established for a long time and its syntax is relatively intuitive, making it possible to roughly predict the resulting color. A drawback of this model is that adjusting the lightness or darkness of a color without changing its hue is nearly impossible by simply modifying its parameters. This makes it difficult to create different shades of the same color in a predictable and manageable way.

```
rgb(45 31 177 / 0.5);
```



Listing 2: Using `rgb()` to define a color from the Kolibri palette with 50% opacity.

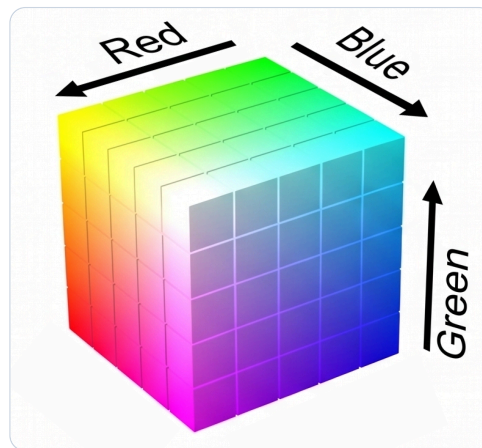


Figure 4: A visual representation of the RGB model with its dimension showcasing how color values are defined. [11]

5.1.2. HSL

CSS Color Module Level 3 introduced the HSL color model to CSS. HSL describes colors using three components: `hue`, `saturation`, and `lightness`, as illustrated in Listing 3 and visually mapped onto the cylindrical coordinate system in Figure 5. The hue is represented as an angle on the color wheel, while saturation and lightness define the intensity and brightness of the color. As with RGB, an optional alpha channel can be added to control opacity.

HSL makes it easier to create variations of a color because saturation and lightness can be adjusted directly without recalculating individual RGB values. This is particularly useful when generating lighter or darker shades while preserving the visual identity of the base color as much as possible. [10]

Although HSL is not as widely used as RGB, it is still commonly used in practice. It is particularly well suited for creating interactive states such as hover effects, as the lightness or saturation of a color can be adjusted independently while largely preserving its hue. Once the distribution of colors around the hue wheel is understood, the resulting colors are also relatively predictable.

```
hsl(256deg 82% 55% / 60%);
```



Listing 3: Using `hsl()` to define a color from the Kolibri palette with 60% opacity.

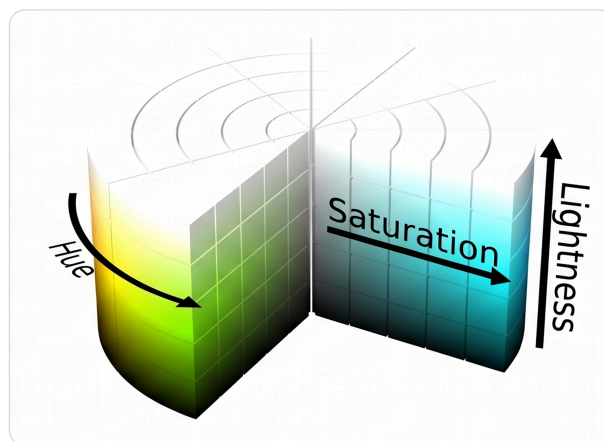


Figure 5: A visual representation of the HSL cylindrical model with its dimensions showcasing how color values are defined. [12]

5.1.3. CIELAB Colors

The Oklab color space and its related Oklch color model make it possible to define colors in a way that more closely matches human vision. They also provide access to a wider range of colors than those that can be represented in the traditional sRGB color space.

The `oklab()` function describes a color using three components: `lightness`, the `a` axis representing red-green variation, and the `b` axis representing yellow-blue variation. The related `oklch()` function uses `lightness`, `chroma`, and `hue` instead. Both functions support an optional alpha channel to control opacity. An exemplary usage of both functions can be observed in Listing 4. While Oklab and Oklch utilize optimized mathematical models for modern screens, they are based on perceptual axes shown in Figure 6, where lightness forms the vertical axis while the color components extend orthogonally from it.

Although this color model is relatively new and not yet widely used, it is regarded as the current state of the art for working with color in CSS because it produces more perceptually uniform results. In practice, this means that adjusting lightness or chroma leads to visual changes that preserve the original base color as accurately as possible compared to older color models. However, its syntax is less intuitive, making it difficult to roughly predict the resulting color without additional knowledge. [10]

```
oklab(63.8% 0.176 -0.279 / 0.7);  
oklch(71.8% 0.255 301.5 / 0.7);
```



Listing 4: Using `oklab()` and `oklch()` to define a color from the Kolibri palette with 70% opacity.

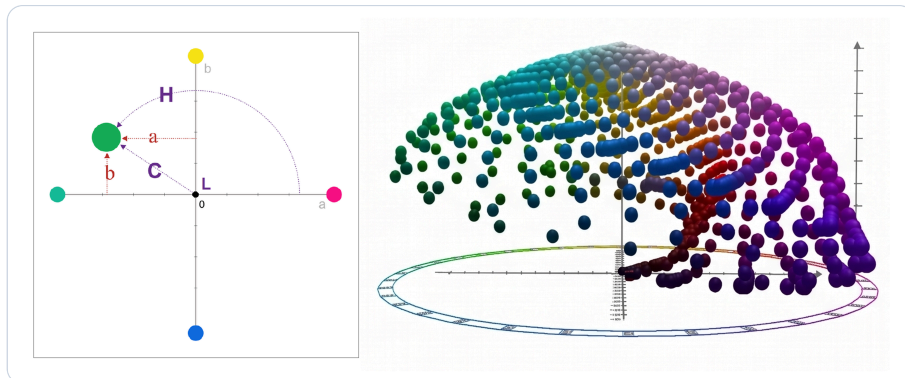


Figure 6: A visual representation of the CIELAB/CIELCH coordinate dimensions, mapping perceptual lightness (L), chromatic axes (a, b), chroma (C), and hue (H). [13], [14]

5.1.4. HWB

The `hwb()` color function was introduced alongside newer CSS color features. It describes colors using three components within the RGB color space: `hue`, `whiteness`, and `blackness`. In practice, it provides an alternative to `hsl()` that can be considered more intuitive when adjusting tints and shades. An example is shown in Listing 5, while Figure 7 illustrates how these coordinates are mapped relative to the peripheral hue wheel. [10]

This color model is not widely used and was designed to provide a simpler and more intuitive approach compared to RGB or HSL. Its simplicity comes from the idea of defining a base color and adding white and/or black to create different shades without changing the underlying hue.

```
hwb(325deg 18% 0% / 30%);
```



Listing 5: Using `hwb()` to define a color from the Kolibri palette with 30% opacity.

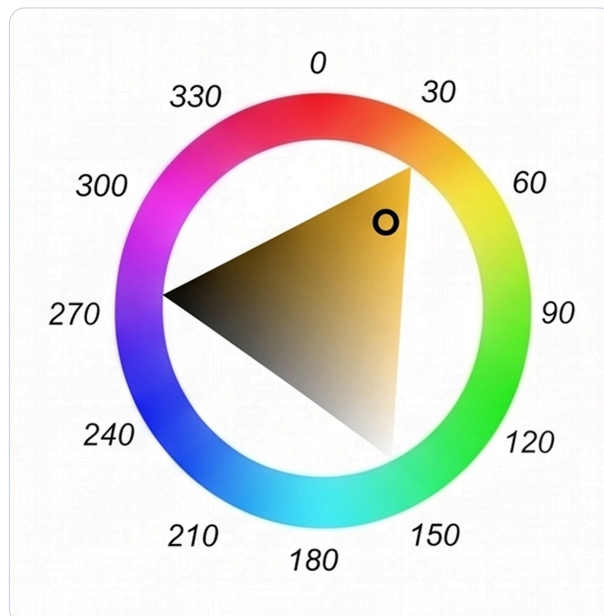


Figure 7: A visual representation of the HWB color space structured as a triangle-wheel picker, showcasing how hue, whiteness, and blackness interact. [15]

5.2. CSS Functions

CSS provides a wide range of built-in functions that simplify common styling tasks. Some of these functions are particularly useful for accessibility. One notable example is `contrast-color()`, which became broadly available during the implementation of this project in April 2026 and is supported by current browser versions. The function returns either black or white, depending on which of the two provides the higher contrast against the color passed as its argument. Listing 6 shows this approach in the context of a `button` element. [16]

```
button {  
  background-color: var(--button-color);  
  color: contrast-color(var(--button-color));  
}
```

Listing 6: Setting the text color of a button to black or white based on the background color.

Using either black or white for text is a reliable way to ensure strong contrast. Depending on the background color, however, the result may appear visually harsh and less pleasant to read. Another CSS function is `light-dark()`, which accepts two color values or images. The first value is used when a light color scheme is active, while the second value is applied when a dark color scheme is active. The active mode is determined by the `color-scheme` CSS property, which indicates the color system an element should render. User agents use this property to adapt elements such as scrollbars, form controls, and the canvas surface to match the preferred scheme. Listing 7 demonstrates this approach using custom properties. [17]

```
body {  
  color: light-dark(var(--color-text--light),  
                   var(--color-text--dark));  
  background-color: light-dark(var(--color-background--light),  
                              var(--color-background--dark));  
}
```

Listing 7: Defining text and background colors based on the active color theme using custom properties.

5.2.1. CSS Custom Functions

Although the growing set of built-in CSS functions covers many use cases, some scenarios, such as calculating `rem` from a pixel value, still require functionality that cannot be expressed with the currently available primitives. In such cases, developers

may need to combine multiple functions or implement custom logic. This is where the experimental `@function` at-rule becomes relevant.

This feature is conceptually similar to CSS custom properties, as custom functions also use names that begin with a dashed-ident `--`. A function is declared using the `@function` keyword, followed by a custom name and an optional list of parameters. Both the function name and its parameters start with `--` and are followed by case-sensitive identifiers defined by the developer. Inside the function body, the expression assigned to `result:` is evaluated and returned, as shown in Listing 8. [18]

```
@function --color-contrast(--color <color>) returns <color> {  
  result: oklch(from var(--color) calc(((1 * -1) + 0.5) * infinity) 0  
  0);  
}
```

Listing 8: A custom function that provides behavior similar to `contrast-color()` and was created before that function became widely available.

As with CSS custom properties, CSS data types such as `<color>`, `<number>`, and `<string>` can be specified for both function parameters and the return value.

5.3. Canvas

The HTML `<canvas>` element is used to draw graphics and animations directly within a web page. By default, the element has no visual representation of its own and must be rendered through JavaScript using either the `Canvas API` or the `WebGL API`. [19]

The `Canvas API` is primarily designed for two-dimensional graphics and is commonly used for tasks such as game rendering, data visualization, image manipulation, and real-time video processing. The `WebGL API` extends these capabilities by providing hardware-accelerated rendering for both two-dimensional and three-dimensional graphics. [20]

In addition to drawing content, the canvas allows pixel data to be read and analyzed. The `ImageData` interface provides access to the underlying RGBA values of a rendered image. Listing 9 demonstrates how the color values of a selected area can be extracted in JavaScript. This is especially useful when converting colors between different CSS color spaces, where manual conversion can introduce inaccuracies. For example, converting an Oklch color into an RGB value requires mapping between different color spaces, which can result in imprecise values that affect further calculations. Using the `canvas` element allows these inaccurate conversions to be avoided by extracting the actual rendered pixel values directly. [21]

```
const canvas = document.createElement('canvas');
const ctx    = canvas.getContext('2d');

ctx.fillStyle = 'oklab(0.638 0.176 -0.279)';
ctx.fillRect(0, 0, 100, 100);

const red    = ctx.getImageData(10, 10, 10, 10).data[0];
const green  = ctx.getImageData(10, 10, 10, 10).data[1];
const blue   = ctx.getImageData(10, 10, 10, 10).data[2];
const alpha  = ctx.getImageData(10, 10, 10, 10).data[3];
```

Listing 9: Creating a purple 100×100 pixel square and extracting the RGBA values from a 10×10 pixel region starting at coordinates $x = 10$ and $y = 10$.

5.4. Contrast Color Function

The previously introduced `contrast-color()` function returns either black or white for a given color, depending on which provides the greater lightness contrast. While this approach is effective, pure black and white can appear visually harsh, especially when used for larger blocks of text.

To address this limitation, an enhanced version of `contrast-color()` was developed during this project and is shown in Listing 10. The custom function accepts a required parameter of type `<color>` and an optional `<percentage>` parameter named `--intensity`.

In the first step, the function determines whether black or white provides the better contrast by evaluating the lightness component of the input color in the Oklch color space. This is achieved by multiplying the lightness value with `-1` and adding `0.5` afterwards, resulting in either a positive or negative value. The result is then multiplied by the constant `infinity`. Due to the way `oklch()` handles lightness values, this effectively collapses the result to either 0 or 1, depending on whether the original color is darker or lighter than 50%. If the lightness value is greater than 50%, black is selected. Otherwise, white is used.

In the second step, the selected contrast color is passed to the `color-mix()` function. The optional `--intensity` parameter controls how much of the original color is mixed back into the result. This produces a softer contrast color that retains some of the visual characteristics of the source color.

To further reduce extreme contrast, the lightness of the selected black or white is adjusted using the `clamp()` function. This ensures that light colors do not exceed a lightness of 97.5% and dark colors do not fall below 15%. As a result, the generated contrast color remains readable while appearing less visually aggressive. The selected bounds represent pragmatic perceptual limits intended to preserve strong readability while reducing the visual harshness associated with maximum contrast combinations.

```

@function --color-contrast(
  --color <color>,
  --intensity <percentage>: 0%
) returns <color> {
  --black-or-white: oklch(
    from var(--color) calc(((1 * -1) + 0.5) * infinity) 0 0
  );

  result: color-mix(
    in oklch,
    oklch(from var(--black-or-white) clamp(0.15, 1, 0.975) c h),
    var(--color) var(--intensity)
  );
}

```

Listing 10: A custom contrast function that softens pure black and white and optionally mixes in a portion of the original color.

Note: The specific thresholds of 15% and 97.5% used in this function serve as a subjective, baseline example to illustrate the concept. These values are not absolute standards and should be modified to align with the specific contrast ratios, aesthetic preferences, and accessibility requirements of the design system in use.

The accompanying example from the project’s example collection also calculates the WCAG 2.x contrast ratio for the generated colors, as shown in Figure 8.

Black or White as Contrast with Optional Color Saturation

Pick a color and get either black or white as contrast color, depending on the color's luminance. Additionally, a saturation percentage can be defined to move from a sharp contrast to a more pleasant one with color added.

Configuration

Background color

Saturation percentage

Results

Background: **#be58fd**

Foreground: **oklch(0.200943 0.024045 354.977)**

Contrast Ratio: **5.19:1**

[Copy Resulting Color](#)

Figure 8: A screenshot of the configuration interface for the custom contrast color function.

To calculate the WCAG 2.x contrast ratio, the RGB values of both colors are required. When colors are defined using functions such as `oklch()`, these values are not directly available, as browsers may preserve the original color format in the computed styles.

A practical workaround is to use the HTML `<canvas>` element. The canvas can be filled with any valid CSS color, and the resulting RGBA values can then be extracted using `getImageData()`. As shown in Listing 11, this technique makes it possible to convert any supported CSS color format into numeric RGBA values.

```
const parseColorToRGBA = (colorString) => {
  const canvas = document.createElement('canvas');
  const ctx    = canvas.getContext('2d');

  ctx.fillStyle = colorString;
  ctx.fillRect(0, 0, 1, 1);

  // Returning data array containing RGBA values as follows: r, g, b, a
  return ctx.getImageData(0, 0, 1, 1).data;
};
```

Listing 11: Extracting the `RGBA` values of any CSS color using the HTML `<canvas>` element.

The implementation of the custom contrast color function is available in the project's repository. [22]

5.5. Ishihara

The Ishihara color test, named after the Japanese ophthalmologist Shinobu Ishihara, was developed in 1917 to detect red-green color vision deficiencies. The test consists of a series of circular plates composed of pseudo-isochromatic dots. These dots vary in size and color and are arranged in patterns that appear distinct to individuals with typical color vision, while blending together for people with certain forms of color blindness. [23]

An example of such a plate is shown in Figure 9. It uses orange and teal tones and is designed to remain visible regardless of the viewer’s color perception. Plates of this kind are commonly used as introductory demonstration images at the beginning of an Ishihara test.

5.5.1. Functional Application in Contrast Testing

Although originally developed as a diagnostic tool, the underlying principle of Ishihara plates also provides a useful framework for evaluating color contrast in interface design.

By placing many small dots of different colors next to each other, the method forces the visual system to rely primarily on chromatic contrast to identify a pattern. In a custom accessibility tool, if the foreground and background colors visually blend together within such a plate, this indicates that the chosen color combination may not provide sufficient perceptual contrast.

Custom plates can be generated using a specific brand palette to verify that key colors remain distinguishable not only for users with typical color vision, but also for users with conditions such as `protanopia` or `deuteranopia`.

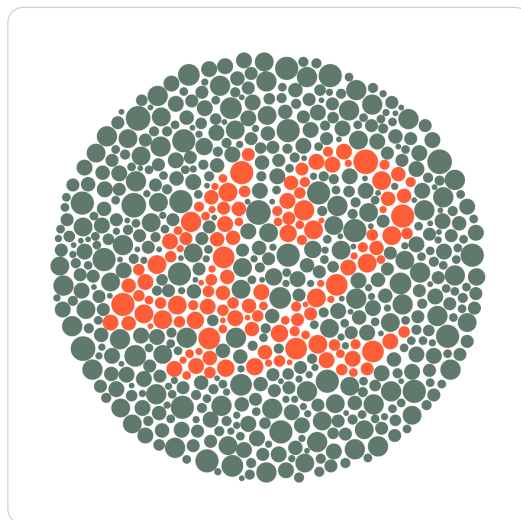


Figure 9: A high-contrast demonstration plate designed to remain legible across all common vision types.

5.6. Interactive Ishihara Plate

The interactive Ishihara plate created during this project is not intended to directly enhance a website through integration into a production environment. Instead, it serves as a developer tool for visually evaluating color contrast in situations where shape, placement, or additional visual cues do not influence recognition. The focus lies entirely on the perception of color itself.

The application allows three separate colors to be defined for the background circles and three additional colors for the foreground circles. Through their arrangement, the foreground circles form the number 42. This setup makes it possible to evaluate how different shades of the same color, for example variations in saturation or lightness, interact with one another and whether sufficient visual contrast remains between foreground and background elements. An example configuration using colors from the Kolibri palette is shown in Figure 10.

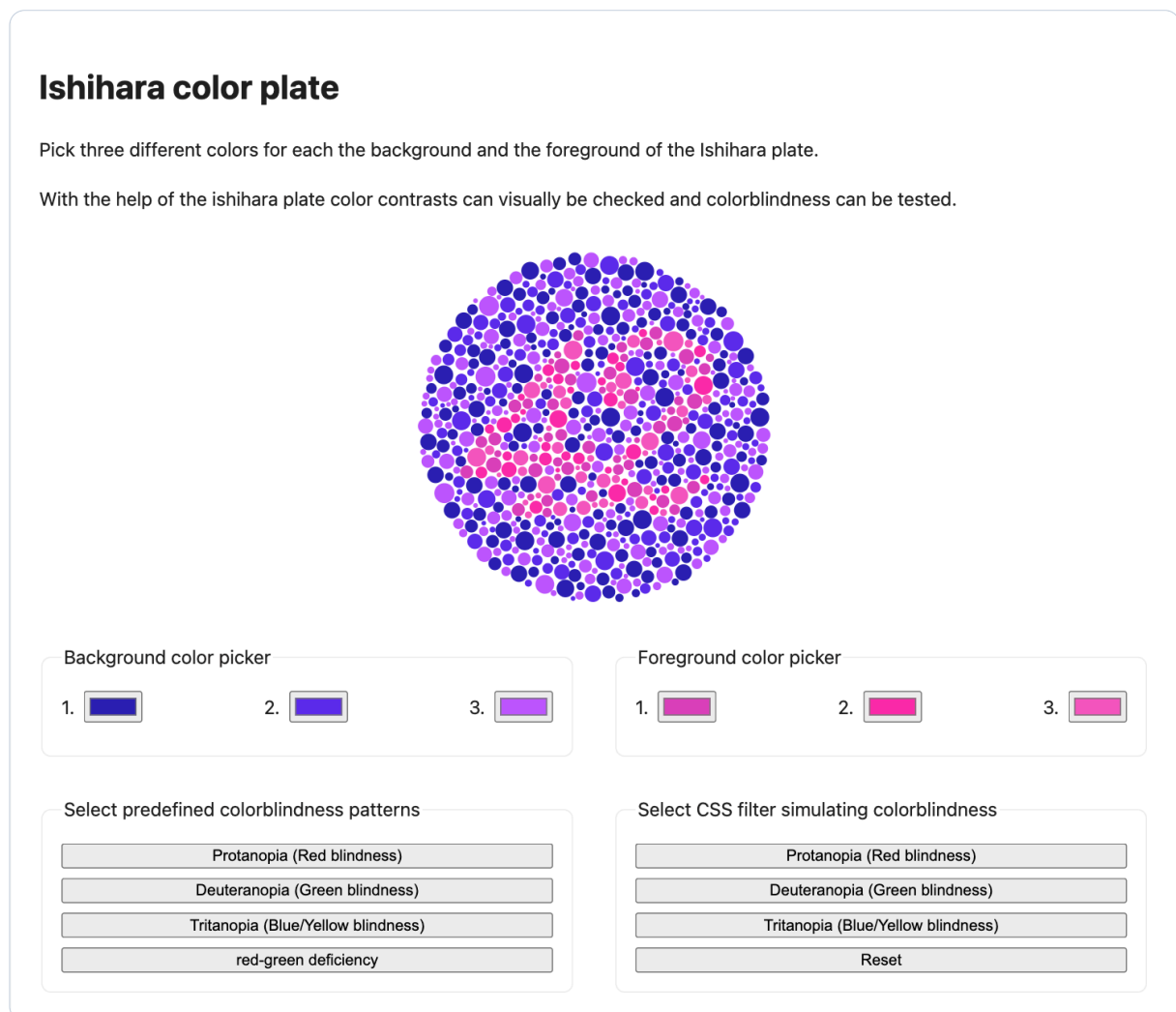


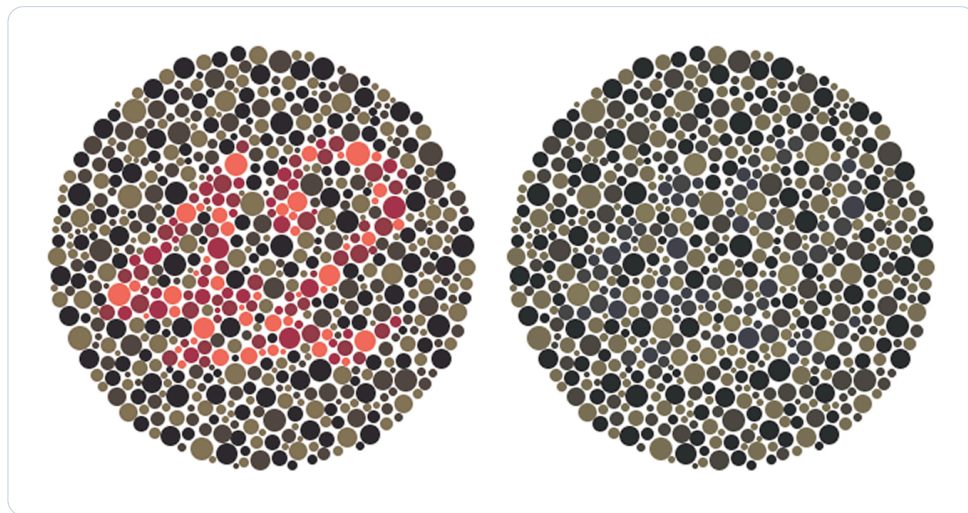
Figure 10: A screenshot of the interactive Ishihara plate configured with colors from the Kolibri palette.

In addition to freely configurable colors, the application also includes predefined color combinations designed to simulate scenarios that are difficult or impossible to distinguish for people with specific forms of color vision deficiency such as **Protanopia**, **Deuteranopia**, and **Tritanopia**. These presets make it possible to evaluate whether certain color combinations remain distinguishable under different forms of impaired color perception.

Although these configurations are inspired by the original Ishihara test plates, the application is not intended to serve as a medically accurate diagnostic tool. Instead, it should be considered a visual indicator that may suggest the need for further professional examination.

An example of these comparison modes can be seen in Figure 11. The left side displays a plate configured with colors that are difficult to distinguish for users with **Deuteranopia**. The right side shows the same plate with a **Deuteranopia** simulation filter applied, illustrating how the color combination may appear to affected users. The simulation filters are based on the bookmarklet filters developed during the previous P7 project.

A more detailed discussion of these bookmarklets is available in the corresponding chapter of the previous P7 project. [8]



*Figure 11: A comparison between a plate configured for **Deuteranopia** on the left and the same plate viewed through a **Deuteranopia** simulation filter on the right.*

The implementation of the interactive Ishihara plate is available in the project's repository. [24]

5.7. CSS Media Queries

CSS Media Queries allow developers to apply different styles based on the characteristics of the device or environment used to display a web page. This mechanism is commonly used to create responsive layouts that adapt their appearance according to the available screen width.

Listing 12 below illustrates the syntax of a media query as defined by the W3C.

```
@media <media-query-list> {  
  <rule-list>  
}
```

Listing 12: Syntax of a media query, where `<media-query-list>` contains `<media-type>` values, `<media-feature>` values, and logical operators.

In web development, the `screen` media type is the most commonly used, but media queries can also target printers by using `print`. If no media type is specified, the default value is `all`, which applies the styles to all devices.

Media features are where media queries become particularly powerful, as more than 40 features are currently available. They allow styles to respond to specific characteristics of the user's environment. Common examples include `max-width` and `max-height`, as well as their counterparts `min-width` and `min-height`. These features are widely used to create responsive websites with layouts optimized for different screen sizes.

Listing 13 demonstrates a media query that uses the `screen` media type together with the `max-width` media feature.

Some media features take accessibility needs and user preferences into account. They make it possible to adapt the presentation of a website to individual requirements. Despite their importance, they are still used less frequently than they should be. The following media features are particularly relevant when building web applications that aim to be accessible to a broader audience. [25]

```
.container {
  display: grid;
  grid-template-columns: 1fr 1fr 1fr;
}

@media screen and (max-width: 768px) {
  .container {
    grid-template-columns: 1fr 1fr;
  }
}
```

Listing 13: Example of a media query that changes the `.container` element layout from 3 columns to 2 columns on smaller screens, such as a vertical tablet screen, using the `screen` media type and the `max-width` media feature.

5.7.1. Reduced Motion

The `prefers-reduced-motion` media feature indicates whether a user has enabled a system setting that reduces non-essential motion. If the value is set to `reduce`, animations and transitions should be limited to the bare minimum.

A value of `no-preference` indicates that the user has not expressed a preference for reduced motion, allowing animations and transitions to be displayed without restriction. [26]

5.7.2. Contrast

The `prefers-contrast` media feature allows users to indicate whether they prefer content to be presented with higher or lower contrast. A value of `no-preference` indicates that the user has not configured a contrast preference.

If the value is set to `more`, the user requests a higher level of contrast. Conversely, a value of `less` indicates that a lower contrast is preferred.

The value `custom` indicates that the user prefers a specific set of colors. This color palette is typically defined through the `forced-colors` media feature, which is explained in the following section. [27]

5.7.3. Forced Colors

The `forced-colors` media feature detects whether the user agent has enabled a forced color mode, such as a Contrast Theme available in the Windows accessibility settings. This media feature has the value `active` when such a mode is enabled and `none` when it is not.

When forced colors mode is active, many color values defined in the style sheet are overridden by the browser during the painting process. The exact colors applied depend on the semantic role of each element. In addition, browsers may draw

backplates behind text to preserve legibility and maintain sufficient contrast when text appears on top of images or patterned backgrounds.

Developers are generally not encouraged to create a separate design specifically for users who enable forced colors mode. Instead, this media query should be used to make targeted adjustments when certain components do not work well in this context. For example, a button that relies solely on `box-shadow` to stand out from the background may lose its visual distinction. In such cases, replacing the shadow with a visible border can provide a more robust solution. [28]

5.7.4. Reduced Transparency

The `prefers-reduced-transparency` media feature indicates whether a user has enabled a system setting to reduce transparent or translucent visual effects. If the value is set to `reduce`, transparency effects such as blurred overlays or frosted glass backgrounds should be minimized or removed. A value of `no-preference` indicates that the user has not expressed a preference and that the default design can be used.

Reducing transparency can improve contrast and readability, particularly for users who find translucent interface elements distracting or difficult to perceive.

Support for this media feature is currently limited and it is not yet fully supported by Firefox and Safari as of July 2026. [29]

5.7.5. Color Scheme

A more widely adopted media feature is `prefers-color-scheme`. It indicates whether the user prefers a light or dark color theme configured through the operating system or the user agent. The value `light` represents a light color theme, while `dark` represents a dark color theme. [30]

5.7.6. Inverted Colors

The `inverted-colors` media feature detects whether the user agent is displaying content with inverted colors. Color inversion can introduce visual side effects, such as shadows appearing as highlights, which may reduce the readability of the content. By responding to this media feature, developers can make targeted adjustments to preserve the visual integrity of the page.

Support for this media feature is currently limited and, at the time of writing, it is only supported by Safari. [31]

5.7.7. Accessing Media Features through JavaScript

The `window` interface provides the `matchMedia()` function, which returns a `MediaQueryList` object. This object can be used to determine whether the current `document` matches a given media query string. Listing 14 demonstrates this approach using the `prefers-color-scheme` media feature.

```
const isDarkTheme = window.matchMedia(  
    '(prefers-color-scheme: dark)'  
).matches;
```

Listing 14: Checking whether the user's system settings prefer a dark color scheme.

This allows developers to respond to user preferences not only in CSS, but also within JavaScript logic and application-specific features. [32]

5.7.8. Changes in the World of Media Queries

CSS media queries continue to evolve, but one important aspect of visual accessibility is still not addressed. Different forms of color blindness and other impairments that affect color perception are not yet covered by an official media feature.

An open issue in the W3C's `csswg-drafts` repository proposes a new media feature called `color-vision-adjustment`. Its goal is to improve web accessibility by allowing developers to respond directly to the needs of users with color vision deficiencies.

The proposal includes support for several specific conditions, including protanopia (red deficiency), deuteranopia (red-green deficiency), tritanopia (blue-yellow deficiency), and achromatopsia (near-total color blindness, resulting in grayscale vision). These impairments were previously investigated in the P7 project through browser bookmarklets designed to simulate the corresponding visual deficiencies.

Further details regarding the implementation and theoretical background of these simulations can be found in the P7 project documentation. [1]

A media feature of this kind would increase awareness of color vision deficiencies and provide developers with a standardized way to react to the specific requirements of affected users. [33]

5.7.9. Deprecated Features

Earlier versions of CSS included media types such as `embossed`, `aural`, and `braille`, which were intended to address specific accessibility needs. These media types were later deprecated because the distinction between device categories was expected to become less meaningful over time. In addition, media types describe classes of devices rather than the actual capabilities they support. [34]

5.7.10. Security Concerns

Information exposed through media queries can be used to build a browser fingerprint, which may allow a device to be identified and tracked across websites. To reduce this risk, browsers may deliberately modify or generalize certain reported values.

Some browsers also provide settings that further limit this information. For example, Firefox offers the “Resist Fingerprinting” option, which causes many media queries to return standardized values instead of device-specific settings. [35]

5.8. CSS Custom Properties

CSS custom properties are user-defined values that can be reused throughout a style sheet where the style defining them is applied. This centralized approach makes styles significantly easier to maintain and reduces code duplication.

A custom property is defined using a `<dashed-ident>`, which consists of two dashes `--` followed by a descriptive identifier. Its value can then be accessed using the CSS `var()` function. Listing 15 demonstrates how the custom property `--primary-color` can be defined once and reused throughout a style sheet. This approach is particularly useful when implementing multiple themes, such as light and dark modes, because colors can be updated from a single central location.

```
body {
  --primary-color: #204ccf;
  --primary-color-contrast: #ffffff;
}

button.primary {
  background-color: var(--primary-color);
  color: var(--primary-color-contrast);
}

h1, h2, h3 {
  color: var(--primary-color);
}
```

Listing 15: Defining a primary color custom property that can be reused throughout the website.

The `@property` at-rule allows custom properties to be defined more precisely. It makes it possible to specify expected syntax definition using `syntax`, whether the property's value is inherited by child elements using `inherits`, and an initial value using `initial-value`. Possible syntax definitions include data type names such as `<color>`, `<number>`, and `<url>`. [36] Listing 16 shows how the previous `--primary-color` example can be expressed using this at-rule. [37]

```

@property --primary-color {
  syntax: '<color>';
  inherits: true;
  initial-value: limegreen;
}

body {
  --primary-color: #204ccf;
}

```

Listing 16: Defining a primary color custom property using the `@property` at-rule.

As shown in Listing 17, custom properties can also be defined in JavaScript using `registerProperty()` or using `setProperty()`. [38], [39]

```

window.CSS.registerProperty({
  name: '--primary-color',
  syntax: '<color>',
  inherits: true,
  initialValue: 'limegreen',
});

document.body.style.setProperty('--primary-color-contrast', '#ffffff');

```

Listing 17: Using `registerProperty()` and `setProperty()` in JavaScript to define CSS custom properties.

Custom properties can also be read in JavaScript using `getPropertyValue()`, as shown in Listing 18. This makes it possible to react to dynamically changed values and use them in application logic.

```

const body      = document.body;
const primaryColor = getComputedStyle(body).getPropertyValue(
  '--primary-color'
);

```

Listing 18: Reading a CSS custom property in JavaScript using `getPropertyValue()`.

5.8.1. Global State Management

CSS custom properties are not limited to storing design values such as colors. They can also be used to manage application state. Because custom properties can be read and modified in both CSS and JavaScript, they provide a convenient way to store configuration values that directly influence the user interface.

For example, a user may not have enabled reduced motion at the operating system level, such as the Reduced Motion setting in the Accessibility settings on macOS, but may choose to disable animations through an application-level preference. This choice can be stored in a custom property such as `--prefers-reduced-motion: true`. The `@container` at-rule can then react to this value and apply different styles at runtime, as shown in Listing 19. [40]

```
:root {  
  --prefers-reduced-motion: false;  
}  
  
@container style(--prefers-reduced-motion: true) {  
  .animated {  
    animation: none;  
  }  
}
```

Listing 19: Using the CSS `@container` at-rule to disable animations based on a custom property.

It is also possible to override custom properties when a different color theme is selected with the `@container` at-rule. For example, the operating system and browser may indicate a preference for a light theme, while the user explicitly selects a dark theme on the website. In this case, a custom property such as `--prefers-dark-theme: true` can be used to switch the application's color palette, as demonstrated in Listing 20.

```
:root {  
  --prefers-dark-theme: false;  
}  
  
body {  
  --text-color:      #222222;  
  --bg-color:        #fcfcfc;  
  --border-color:    #eeeeee;  
  --primary-accent:  #0056b3;  
}  
  
@container style(--prefers-dark-theme: true) {  
  body {  
    --text-color:      #ededed;  
    --bg-color:        #212121;  
    --border-color:    #232323;  
    --primary-accent:  #6Db3f4;  
  }  
}
```

Listing 20: Using the `@container` at-rule to switch to a dark color theme based on a custom property.

5.8.2. Custom Property Toggle

In modern CSS, developers often face the challenge of managing repetitive declarations, especially when implementing light and dark themes. Whether the switch is handled through classes or through

`@media (prefers-color-scheme: dark)`, the same property names typically need to be declared multiple times with different values. A lesser-known detail in the CSS specification makes it possible to implement this kind of state management more elegantly by using custom properties as toggle values.

According to the CSS specification, a custom property must contain at least one token to be considered valid. A single whitespace character satisfies this requirement and can therefore be used to simulate boolean-like values, as shown in Listing 21. [41]

```
body {  
  --ON: ;  
  --OFF: initial;  
}
```

Listing 21: Both `' '` and `initial` are valid values for custom properties.

In traditional theming setups, each variable must be defined separately for both the light and dark theme. By using the toggle values introduced in Listing 21, the theme state can be stored in a single custom property, as demonstrated in Listing 22.

```
body {  
  --is-light-theme: var(--ON);  
}  
  
@media (prefers-color-scheme: dark) {  
  body {  
    --is-light-theme: var(--OFF);  
  }  
}
```

Listing 22: Assigning either `' '` or `initial` depending on the active theme creates the basis for the toggle logic.

When `--is-light-theme` is used as a prefix in another custom property declaration, the resulting declaration is either valid or invalid depending on its value. If the prefix expands to a whitespace character, the declaration remains valid. If it expands to `initial`, the declaration becomes invalid. To handle this, the `var()` function accepts a fallback value as a second argument, which is automatically applied if the

referenced custom property resolves to a guaranteed-invalid value like `initial`. Listing 23 illustrates how this mechanism is utilized. [42]

```
body {
  --color-text--light: var(--is-light-theme) black;
  --color-text--dark: white;

  --color-background--light: var(--is-light-theme) white;
  --color-background--dark: black;

  --color-text: var(--color-text--light,
                    var(--color-text--dark));
  --color-background: var(--color-background--light,
                          var(--color-background--dark));
}
```

Listing 23: When a custom property becomes invalid, `var()` automatically uses the fallback value.

Multiple fallback values based on different toggle properties can be chained using the CSS `var()` function. [43] This enables centralized state management, where the custom properties used throughout the entire style sheet are controlled from a single location. [44]

An example of this theme toggle is available in the project’s repository. [45]

6. Empowering Users to Refine Their Experience

Lastly, the focus shifts from developers to the users interacting with web applications. Developers are able to optimize many aspects of a website based on assumptions about user preferences, for example through media queries. However, enabling users to further refine and personalize their own experience allows for a greater level of individual adaptation. This chapter focuses on a more powerful version of the previously introduced Developer Widget, providing users with the necessary tools to customize their experience for each web application individually.

6.1. View Transition API

The View Transition API provides a native mechanism for animating transitions between different states of a web application. It can be used to create smooth visual effects when navigating between pages or when updating views within single-page applications (SPAs).

For transitions that occur within the same document, the state change is wrapped inside the `document.startViewTransition()` method. Cross-document transitions in multi-page applications (MPAs) are triggered automatically during navigation and can be enabled through the `@view-transition` at-rule. Internally, the API captures snapshots of both the previous and the new state and animates between them. By default, this transition consists of a simple fade effect, where the old view gradually decreases its opacity to `0` while the new view increases its opacity to `1`.

The generated animations can be customized through the `::view-transition-old()` and `::view-transition-new()` pseudo-elements, which target the previous and new view respectively. Shared styling can be applied through `::view-transition-group()`. These pseudo-elements accept different targets, including `root` for animating the entire page, `*` for all transition targets, or a custom `view-transition-name` to animate specific elements independently.

An example of a custom page transition is shown in Listing 24. In this case, the previous page is animated out of view towards the left while the new page enters from the right, creating a horizontal swipe effect. [46]

```

@keyframes move-out {
  from {
    transform: translateX(0%);
  }
  to {
    transform: translateX(-100%);
  }
}

@keyframes move-in {
  from {
    transform: translateX(100%);
  }
  to {
    transform: translateX(0%);
  }
}

::view-transition-old(root) {
  animation: move-out 0.4s ease-in both;
}

::view-transition-new(root) {
  animation: move-in 0.4s ease-in both;
}

```

Listing 24: A custom page transition that swipes the previous view out to the left while moving the new view in from the right.

6.2. Validation and Verification

The Preferences Widget underwent smaller-scale user testing to gather feedback from real users and refine the interface for future integration into Kolibri. The tests were intentionally kept simple to allow participation from users with different backgrounds and experience levels. The primary focus was usability, keyboard accessibility, intuitiveness, and the overall visual experience.

The documentation and results of the conducted user tests are available in the project’s repository under the usertests section. [47]

The different example applications and prototype pages were additionally validated using automated accessibility testing tools such as the axe DevTools browser extension and Google Lighthouse. These automated checks were supplemented with manual keyboard accessibility testing to verify practical usability beyond purely automated evaluation.

6.3. Preferences Widget

CSS media queries already make it possible to respond to certain user preferences. However, this only works when the user has configured the corresponding settings in the operating system or browser. If no such settings exist, websites have no reliable way to adapt to the user's individual accessibility needs.

Even when system-level preferences are configured, they may not reflect the requirements of every website. A user may prefer reduced motion in general, but still want to enable animations on a specific page, or the opposite. In addition, some accessibility preferences, such as color blindness simulations, are not currently covered by CSS media queries and therefore cannot be addressed through native browser mechanisms alone.

To overcome these limitations and give users greater control, the preferences widget shown in Figure 12 was developed. It uses existing system settings as a starting point, while allowing them to be adjusted and extended on a per-website basis.

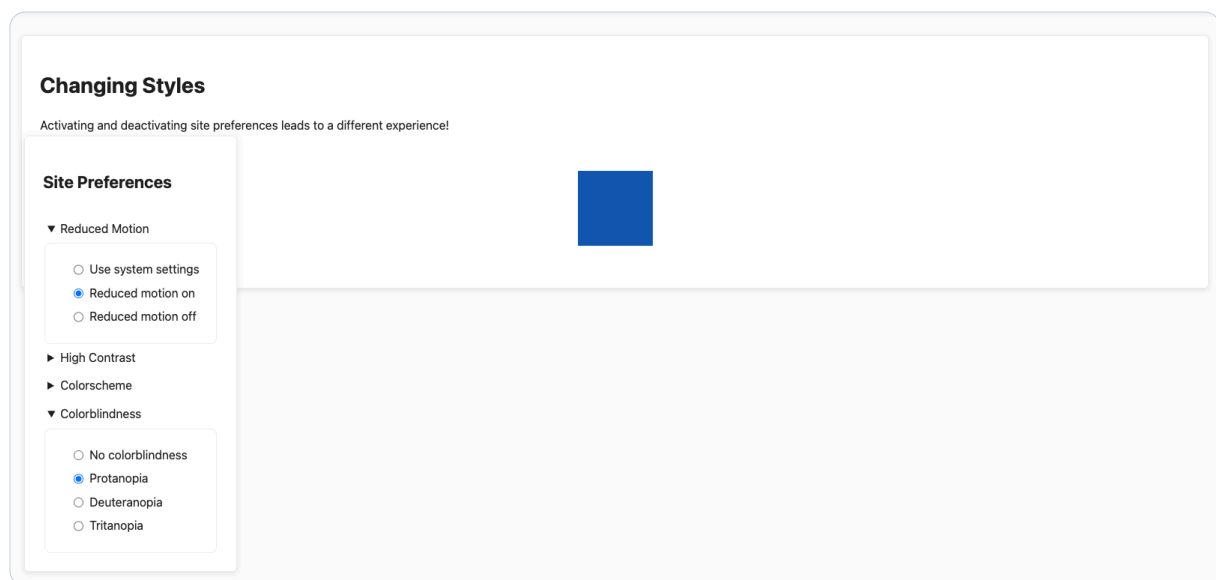


Figure 12: A screenshot of the preferences widget with page-specific settings for reduced motion and color blindness.

6.3.1. JavaScript Part

Where supported, existing CSS media queries serve as the foundation for the widget's default values. When a user visits the page for the first time, the current operating system or browser settings are used as the initial configuration. Any changes made through the widget are stored in the browser's `localStorage`, ensuring that preferences persist across page reloads and remain available throughout the application.

Each preference is also written to a CSS custom property on the `root` element, making the value accessible from both JavaScript and CSS. This approach establishes a single source of truth that can be used consistently across the entire application, as shown in Listing 25.

```
const setAccessibilityProperty = (property,
                                value,
                                mediaQueryString) => {

  localStorage.setItem(property, value);

  if(value === 'system' && mediaQueryString !== '') {
    const systemSetting = window.matchMedia(mediaQueryString).matches;
    document.documentElement.style.setProperty(property,
systemSetting);
  } else {
    document.documentElement.style.setProperty(property, value);
  }
}
```

Listing 25: The `setAccessibilityProperty()` function stores a value in `localStorage` and updates the corresponding custom property on `root`.

The `getAccessibilityProperty()` function shown in Listing 26 reads a previously saved value from `localStorage`. If no value has been stored yet, it returns `'system'`, indicating that the operating system or browser setting should be used as the default.

```
const getAccessibilityProperty = (property) => {
  const savedProperty = localStorage.getItem(property);

  if(savedProperty) {
    return savedProperty;
  } else {
    return 'system';
  }
}
```

Listing 26: The `getAccessibilityProperty()` function returns a saved value or falls back to the system setting.

During page initialization, both functions are used to populate the custom properties with either the stored value or the corresponding system preference, as shown in Listing 27.

```

setAccessibilityProperty('--prefers-reduced-motion',
    getAccessibilityProperty(
        '--prefers-reduced-motion'),
    '(prefers-reduced-motion: reduce)');
setAccessibilityProperty('--prefers-contrast',
    getAccessibilityProperty(
        '--prefers-contrast'),
    '(prefers-contrast: more)');
setAccessibilityProperty('--prefers-dark-theme',
    getAccessibilityProperty(
        '--prefers-dark-theme'),
    '(prefers-color-scheme: dark)');
setAccessibilityProperty('--prefers-colorblind-mode',
    getAccessibilityProperty(
        '--prefers-colorblind-mode'),
    '');

```

Listing 27: Initializing the global custom properties with stored values or system defaults.

The `syncWidgetOption()` function shown in Listing 28 synchronizes the values stored in the custom properties with the corresponding radio buttons in the widget. By reading the values directly from `root`, the custom properties remain the single source of truth for both the user interface and the underlying logic.

```

const syncWidgetOption = (option, property) => {
    const value = getAccessibilityProperty(property);

    option.forEach((radio) => {
        radio.checked = radio.value === value;
    });
}

syncWidgetOption(motion, '--prefers-reduced-motion');
syncWidgetOption(contrast, '--prefers-contrast');
syncWidgetOption(colorscheme, '--prefers-dark-theme');
syncWidgetOption(colorblindness, '--prefers-colorblind-mode');

```

Listing 28: Synchronizing the selected radio buttons with the values stored in the custom properties.

Finally, each radio group registers a `change` event listener that updates both `localStorage` and the corresponding custom property whenever the user selects a different option, as shown in Listing 29.

```

addOptionEventListener(motion,
    '--prefers-reduced-motion',
    '(prefers-reduced-motion: reduce)');
addOptionEventListener(contrast,
    '--prefers-contrast',
    '(prefers-contrast: more)');
addOptionEventListener(colorscheme,
    '--prefers-dark-theme',
    '(prefers-color-scheme: dark)');
addOptionEventListener(colorblindness,
    '--prefers-colorblind-mode',
    '');

const addOptionEventListener = (option, property, mediaQueryString) =>
{
    option.forEach((radio) => {
        radio.addEventListener('change', () => {
            if(radio.checked) {
                setAccessibilityProperty(
                    property, radio.value, mediaQueryString
                );
            }
        });
    });
};

```

Listing 29: Updating the user's preference when the selected radio button changes.

6.3.2. CSS Part

Once the CSS custom properties representing the user's preferences are in place, they can be used to create different visual representations of the website. In Listing 30, a set of custom properties defines the default styling. These values are selectively overridden using the CSS `@container` at-rule when the user prefers a dark color scheme.

```

body {
  --bg-color:      #fcfcfc;
  --text-color:    #222222;
  --content-bg:    #ffffff;
  --border-color:  #eeeeee;
  --primary-accent: #0056b3;
  --shadow:        0 2px 8px rgba(0, 0, 0, 0.1);

  color-scheme: light;
}

@container style(--prefers-dark-theme: true) {
  body {
    --bg-color:      #212121;
    --text-color:    #ededed;
    --content-bg:    #2e2e2e;
    --border-color:  #232323;
    --primary-accent: #6db3f4;
    --shadow:        0 2px 8px rgba(0, 0, 0, 0.5);

    color-scheme: dark;
  }
}

```

Listing 30: Overriding the default custom properties with the CSS `@container` at-rule on `body` to apply a dark color scheme.

As additional preferences are introduced, the number of possible combinations increases. Listing 31 demonstrates how the base styling is adjusted when both the dark theme and the high contrast mode are enabled, resulting in a different visual presentation.

```

@container style(--prefers-dark-theme: true) and
style(--prefers-contrast: true) {
  body {
    --bg-color:      #000000;
    --text-color:    #ffffff;
    --content-bg:    #000000;
    --border-color:  #ffffff;
    --primary-accent: #80c5ff;
    --shadow:        none;
  }
}

```

Listing 31: Applying a dedicated style when both dark mode and high contrast mode are enabled.

In some cases, changing individual custom property values is not sufficient and additional CSS rules must be applied. The `@container` at-rule combined with the `style()` query makes this possible. Because the preference properties are defined on `body`, they can be used to conditionally apply styles across the entire application. Listing 32 demonstrates how animations and transitions can be disabled when the user prefers reduced motion.

```

@container style(--prefers-reduced-motion: true) {
  .animated-contents {
    animation: none !important;
  }

  .transitional-contents {
    transition: none !important;
  }
}

```

Listing 32: Applying additional styles based on custom properties using the `@container` rule and a `style()` query.

6.3.3. The Paradox of Reduced Motion Transitions

An interesting and somewhat ironic aspect is that enabling reduced motion preferences does not necessarily mean that all transitions and animations should be removed from a webpage. In certain contexts, the opposite can even be beneficial. Different accessibility concerns require different solutions, and some of these measures may appear counterintuitive at first glance.

For example, users who prefer reduced motion may still benefit from smooth and controlled transitions when switching between color themes. Abrupt flashes or

immediate color changes, even when triggered intentionally by the user, can create discomfort or visual strain. In such situations, introducing subtle transitions may improve the overall experience rather than worsen it.

Although it may initially seem contradictory to apply transitions when `--prefers-reduced-motion: true` is configured, carefully controlled color transitions can therefore represent a valid accessibility measure.

6.3.3.1. Animating Properties

To animate CSS custom properties, the `@property` at-rule is required. This informs the browser about the expected value type of a property, such as `<color>`, allowing it to interpolate the values during transitions. Once defined, transitions can be applied directly to the custom properties within a `@container` rule using the global `*` selector, since the registered properties themselves are not scoped locally. Listing 33 demonstrates how custom color properties can be registered and transitioned.

```
@property --bg-color {
  syntax: '<color>';
  inherits: true;
  initial-value: #f0f0f0;
}

@property --text-color {
  syntax: '<color>';
  inherits: true;
  initial-value: #222222;
}

@container style(--prefers-reduced-motion: true) {
  * {
    transition: --bg-color 0.3s linear,
               --text-color 0.3s linear,
  }
}
```

Listing 33: Having custom properties defined with `@property` to be able to add a transition within a `@container` rule.

6.3.3.2. View Transition

While animating individual properties can be useful, it may result in less coherent transitions when multiple properties change simultaneously. In such situations, individual animations can compete for attention and create a visually noisy effect that is ultimately more distracting than beneficial. To avoid this, it is often preferable to transition the page state as a whole by using the View Transition API.

For the Preferences Widget, this can be achieved by wrapping the property update inside the `document.startViewTransition()` method, as shown in Listing 34. This allows the browser to capture the previous and new state of the page and animate the transition between them in a coordinated manner. The resulting animation can be further refined through CSS. For example, the duration can be increased to create a smoother transition, as demonstrated in Listing 35.

```
document.startViewTransition(() => {  
  document.documentElement.style.setProperty(property, appliedValue);  
});
```

Listing 34: Enabling a view transition for a property change using the `startViewTransition()` method.

```
::view-transition-group(root) {  
  animation-duration: 0.6s;  
}
```

Listing 35: Increasing the duration of the view transition to create a smoother visual effect.

The implementation of the preference widget, including several example settings that demonstrate how individual and combined preferences affect the website's styling, is available in the project's repository. [48]

7. Discussion

Generally speaking, modern CSS and JavaScript already provide powerful accessibility primitives, and many accessibility concerns do not require large or complex frameworks to be addressed effectively. Native capabilities are often overlooked or underestimated by developers, either due to a lack of awareness or because many modern projects begin directly with frameworks instead of first considering native approaches. In some cases, developers entering the industry are barely exposed to native environments at all, as frameworks are already introduced during apprenticeships before the fundamentals of the web platform itself are fully understood.

Frameworks certainly provide quality-of-life improvements and can simplify development in many scenarios. However, in some areas, avoiding unnecessary abstractions can reduce complexity and improve transparency. Working with native technologies and minimizing dependencies often allows for more direct customization and better adaptation to individual accessibility needs.

It is also remarkable how quickly browser standards evolve. During the timespan of this project, for example, the `contrast-color()` CSS function transitioned from limited browser support to broad support across all major browsers.

At the same time, browser support for CSS and JavaScript features still contains inconsistencies. Certain areas feel incomplete or missing fundamental capabilities. However, these shortcomings should also be viewed in the context of the web platform's continuous development, as existing gaps are gradually closed while new requirements emerge over time.

One example is the extraction of `RGB` values from `oklch()` colors. At the time this project was written, this was not directly possible in a practical way. A reasonable expectation would be for `getComputedStyle()` to return computed `RGB` values even when colors are defined using `oklch()`. Instead, the original `oklch()` value is returned unchanged. To calculate contrast ratios, the implementation therefore had to rely on the HTML `<canvas>` element to extract the required `RGBA` values through `getImageData()`.

Another particularly interesting aspect was working with the experimental `@function` at-rule. Since the feature is still experimental, incomplete behavior is expected to some extent. However, the biggest challenge was the lack of debugging possibilities combined with sparse and partially vague documentation. Developing with this feature often became a trial-and-error process because there was no reliable way to inspect intermediate states during calculations or identify why certain expressions failed. Some supported behaviors are only loosely documented, while other surprisingly useful capabilities are not documented at all despite working reliably in practice.

An additionally challenging aspect is passing relevant intermediate values into calculations. Functions such as `oklch()` allow access to properties like the lightness component of a color, and these values can be manipulated directly within the function itself. However, there is currently no practical way to extract such intermediate values into custom properties for further processing. This makes it difficult to build more complex conditional logic, for example when trying to dynamically alter output values with the `if()` function based on calculated lightness or other color characteristics.

The current binary pass/fail nature of the WCAG 2.x contrast requirements is another aspect worth discussing. The calculations are heavily simplified and even contain known issues, such as the threshold value `0.03928` being used instead of `0.04045`, which originated from a rounding error in the original WCAG draft and later became fixed within the specification. Furthermore, the current model does not properly account for perceptual factors. A contrast ratio may technically satisfy the guideline while still appearing visually uncomfortable or less readable in practice.

This is where APCA introduces promising improvements. The model considers perceptual aspects such as text size, font weight, and surrounding context, acknowledging that readability changes depending on how content is visually presented. Once further refined and standardized, this approach could provide a far more realistic foundation for accessibility contrast evaluation across the web.

8. Conclusions

Improving accessibility and usability for a diverse group of users does not necessarily require large or complex solutions. This project demonstrates that existing web standards can already be extended with relatively little implementation effort to provide a more user-centric and customizable experience for how web content is presented.

The Preferences Widget in particular shows how native browser capabilities and CSS custom properties can be combined to create flexible accessibility configurations on a website-specific level. At the same time, this approach introduces an important challenge. The more configuration options become available, the more visual combinations and states must be considered, implemented, and tested. Combining multiple accessibility preferences quickly results in a large number of possible variations. This complexity represents the biggest limitation of such an approach and makes it difficult to realistically cover every possible combination. In practice, implementations therefore need to focus on the settings that provide the greatest benefit for the broadest group of users.

8.1. Limitations

The scope of this project was limited and therefore no large-scale user testing was conducted to quantitatively validate the assumptions and implementations presented throughout this documentation. In addition, perceived contrast remains partially subjective. While contrast calculations and perceptual models can improve accessibility for many users, no universal solution exists that guarantees optimal readability and visual comfort for everyone.

APCA represents a significant improvement over the current relative luminance approach used in WCAG 2.x because it considers perceptual aspects such as text size, font weight, and surrounding context. However, the underlying theory and calculations are considerably more complex than the current WCAG 2.x model. Furthermore, APCA is still under active development and has not yet been finalized or officially established as a web standard. As a result, it is still too early to draw definitive conclusions regarding its long-term adoption and practical impact.

8.2. Future Work

In its current form, the Preferences Widget cannot yet be integrated directly into the Kolibri Web UI toolkit. Although this was not part of the original project scope, the widget underwent user testing to evaluate its usability and effectiveness. Based on the insights gained during these tests, a draft proposal for integration into Kolibri was developed. The draft implementation can be found in the following repository. [49]

An exemplary implementation of the widget within the Kolibri ecosystem is also available in the same repository under the examples section. [50]

The example application for the custom contrast color function could also be extended with APCA support in the future. This would make it possible to evaluate not only WCAG 2.x contrast requirements but also the future WCAG 3.0 perceptual contrast model once the specification becomes finalized and publicly established.

Another valuable next step would be conducting a survey focused on identifying the most important accessibility customization needs users expect from modern websites. Such research could help determine which configuration options provide the greatest practical value for the initial implementation of the Preferences Widget and which accessibility preferences are most relevant in everyday web usage.

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List of Aids

Artificial Intelligence and Language Models

Tool: ChatGPT (OpenAI), Version GPT-5.5.

Type of Usage: Used for linguistic refinement of drafts and improvement of readability and text flow.

Extent: Specific sections of the documentation were refined using the tool to improve wording, transitions, and overall readability. The original content, technical facts, code, and project structure remained author-created and were manually verified for accuracy.

Tool: Gemini (Google), Version 3 Flash.

Type of Usage: Used for linguistic refinement of drafts and formating mathematical formulas (Relative Luminance).

Extent: Specific paragraphs regarding WCAG luminance calculations were refined using the tool. All technical facts and code were manually verified for accuracy.

Tool: Gemini (Google), Version 3 Flash.

Type of Usage: Used for graphic editing and uniform styling of figures.

Extent: The backgrounds of the three-dimensional color model illustrations were mechanically modified to a uniform white color for visual consistency. The original geometric structures, annotations, and colors remained completely unaltered.

MSE-Declaration of Originality

The project reports (P7 and P8) and the master's thesis (P9) must be accompanied by the following declaration of originality and signed:

I hereby declare that any individual work submitted for assessment is entirely the product of my own effort:

- That I have correctly cited all text passages that do not originate from me, in accordance with standard academic citation rules (e.g., APA or IEEE), and that I have clearly mentioned all sources used;
- That I have declared in footnotes or in an "List of Aids" all aids used (AI assistance systems such as chatbots [e.g., ChatGPT], translation [e.g., DeepL], paraphrasing [e.g., QuillBot]) or programming applications [e.g., GitHub Copilot] and indicated their use at the corresponding text passages;
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Florian Schnidrig

First Name / Surname

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